The Vacuum Is an Area Law. Negative Energy Is Not Λ.

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Abstract

Paper 56 closed occupation-transfer Λ: every swap raises $\langle H \rangle$. The unperturbed Fermi sea has $E_{\mathrm{vac}} < 0$ (open -1657, periodic -1723). If that energy were a first-law volume, Gauss would give outward $a \propto r$.

It is not a first-law volume. Vacuum S tracks cut area ($\rho = 0.9999$), not ball volume. $S(5)/S(2) = 8.11$ sits next to $A(5)/A(2) = 6.23$, not $V(5)/V(2) = 15.6$. The shape $a_{\mathrm{try}} = -S/R^2$ is nearly constant (slope 0.26), not a sea.

Auditor, $N = 10$: $E_{\mathrm{vac}} < 0$ CONFIRMED. Area over volume CONFIRMED. E_{vac} moves with N (-950 on the auditor). That is a cutoff Fermi-sea energy, not a measured cosmological constant.

Admission. *I looked at the vacuum. The energy is negative. The entropy is an area. I did not call E_{vac} de Sitter.*